MILITARY SPECIFICATION SHEET

(B) CABLE ASSEMBLY, RADIO FREQUENCY (AIRCRAFT AUDIO), CX-12987/AR

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of specification MIL-C-22442.

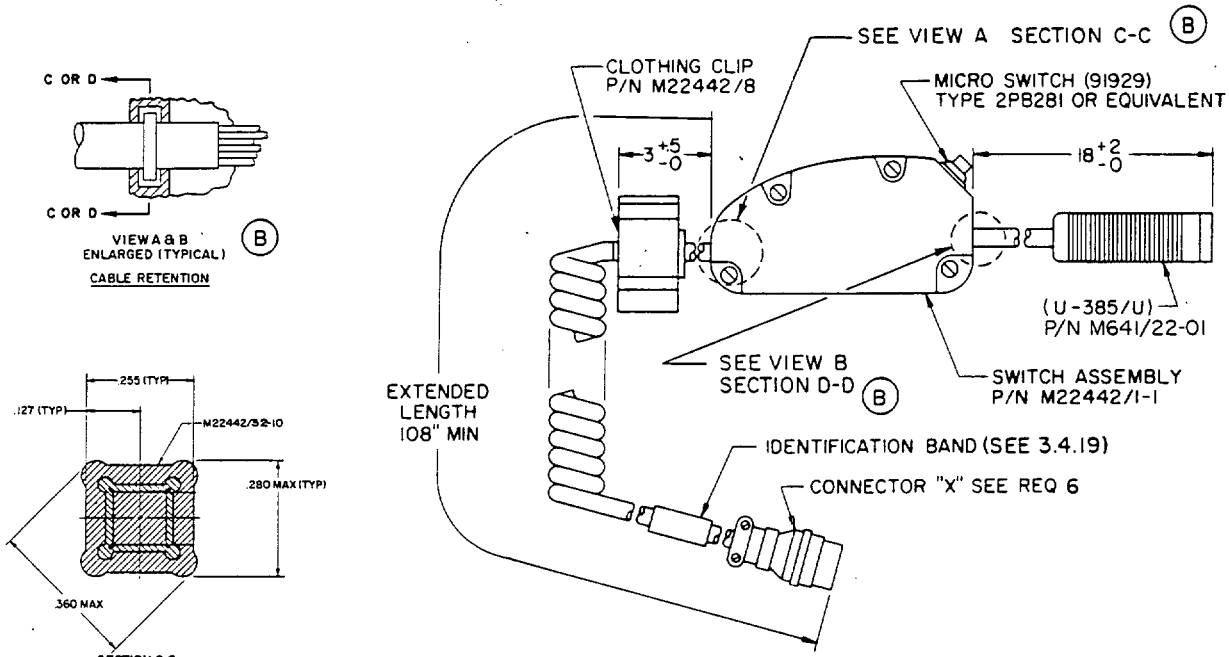
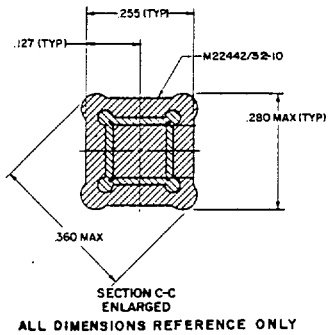
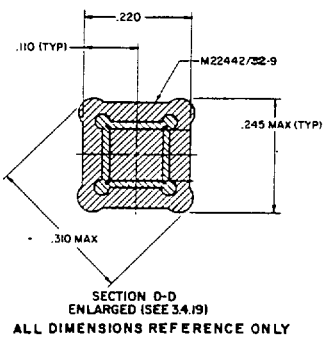
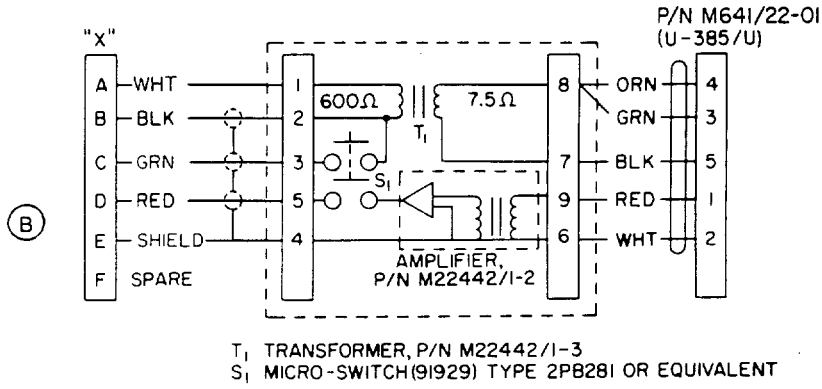


FIGURE 1. CABLE ASSEMBLY CX-12987/AR
P/N M22442/22-1







T_1 TRANSFORMER, P/N M22442/1-3
S_1 MICRO-SWITCH (91929) TYPE 2PB281 OR EQUIVALENT

FIGURE 2. CABLE ASSEMBLY CX-12987/AR
P/N M22442/22-1 SCHEMATIC DIAGRAM

AMSC N/A FSC 5995

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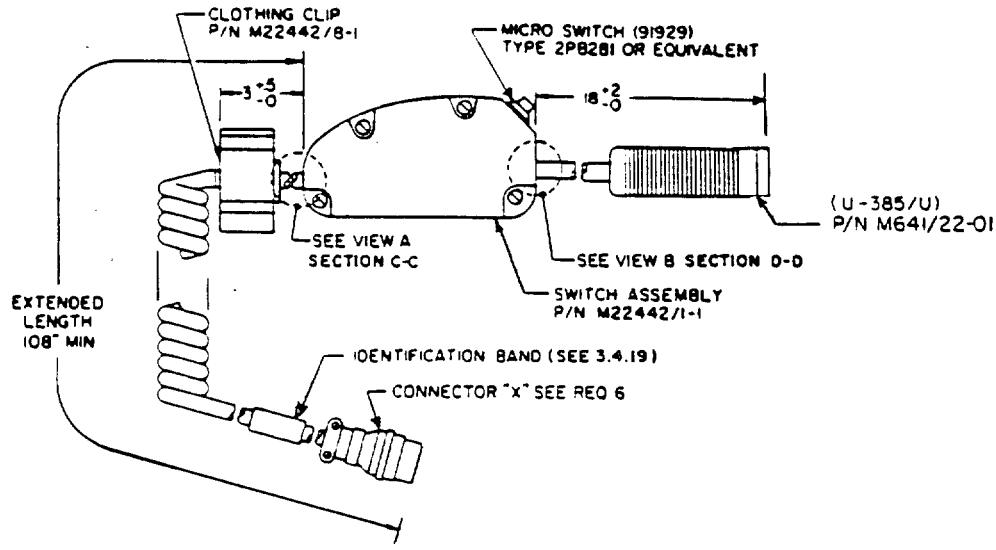


FIGURE 3. Cable assembly P/N M22442/22-2.

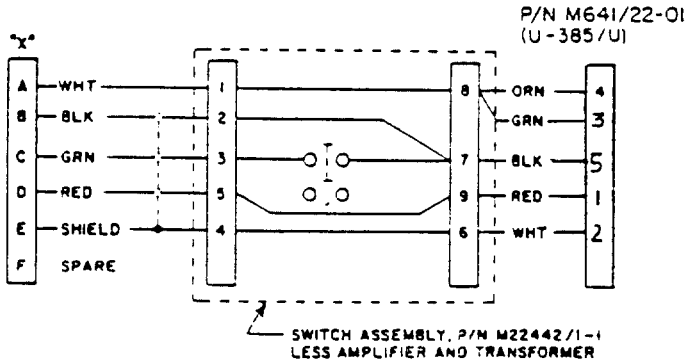


FIGURE 4. Cable assembly P/N M22442/22-2 schematic diagram.

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APPLICABLE DOCUMENTS:

MIL-C-10392	Cord, Electrical (Audio, Miniature)
MIL-J-641/22	Jacks, Telephone, General Specification for
MS3116	Connector, Plug, Electric

REQUIREMENTS:

1. The cable between the switch assembly and the "X" connector shall be in accordance with MIL-C-22442/3. The cable between the switch assembly and the P/N M641/22-01 (U-385/U) connector shall be in accordance with MIL-C-10392 Type WM-59A/U.

(B) 2. The weight of the cable assembly shall not be greater than 20 ounces.

3. The stay cord shall be firmly attached to the connectors and the switch assembly.

(B) 4. The amplifier shall be in accordance with M22442/1-2.

(B) 5. The transformer shall be in accordance with M22442/1-3.

(B) 6. Connector "X" shall be MS3116F10-6P or functional equivalent. Black tubing may be used as a bushing under strain relief.

7. The DC resistance of the M22442/22-1 cable assembly shall be as indicated when tested with a suitable ohmmeter.

(B) (a) Contact A of "X" and contact B of "X" less than 100 ohms.

(B) (b) Contact A of "X" and contact C of "X" with the switch depressed less than 100 ohms.

(B) (c) Contact E of "X" and contact 2 of P/N M641/22-01 (U-385/U) less than 4 ohms.

(B) (d) Contact 3 of P/N M641/22-01 (U385/U) and contact 5 of P/N M641/22-01 (U385/U) less than 10 ohms.

(B) (e) Contact 4 of P/N M641/22-01 (U385/U) and contact 5 of P/N M641/22-01 (U385/U) less than 10 ohms.

8. The cable assembly shall be tested in accordance with the applicable tests listed in Table I.

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9. The DC resistance of the M22442/22-2 cable assembly shall be as indicated when tested with a suitable ohmmeter.

   (a) Contact A of "X" and contact 4 of P/N M641/22-01 (U-385/U) less than 4 ohms.

   (b) Contact B of "X" and contact 5 of P/N M641/22-01 (U-385/U) less than 4 ohms.

   (c) Contact C of "X" and contact 5 of P/N M641/22-01 (U-385/U) less than 4 ohms when the switch is depressed.

   (d) Contact D of "X" and contact 1 of P/N M641/22-01 (U-385/U) less than 4 ohms.

   (e) Contact E of "X" and contact 2 of P/N M641/22-01 (U-385/U) less than 4 ohms.

NOTES:

1. All dimensions are in inches.

2. Tolerances are metal parts $\pm$.005, molded sections $\pm$.05 inches unless otherwise specified.

3. MIL-C-81867 is superseded by MIL-C-22442/22.

Preparing Activity:
Navy - AS
(Project No. 5995-N118-5)

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(B) TABLE I.

Inspection	Requirement Paragraph	Test Paragraph	Group I Qualification Tests	Group II Quality Conformance Test 100%	Group III Sample Testing Per 4.5.1
GROUP A (ALL SAMPLE UNITS) 1/					
Visual & Mechanical Inspection	3.1 & 3.4	4.6.2	X	X	
Soldering	3.4.24	4.6.2.1	X	X	
Transformer Efficiency 2/	5	10a	X		X
Transformer Input Matching Impedance 2/	6	10b	X		X
Transformer Distortion 2/	7	10c	X		X
Transformer dc Winding Resistance 2/	8	10d	X		X
Transformer Output 2/	9	10e	X	X	
Amplifier Frequency Response 2/	13	19b	X	X	
Amplifier Input Power 2/	14	19c	X		X
Amplifier Distortion 2/	15	19d	X		X
Amplifier Linearity 2/	16	19b	X		X
Amplifier Stability 2/	17	19e	X		X
Amplifier Input Impedance 2/	18	19f	X		X
Continuity	3.5.3	4.6.5	X	X	

1/ Minimum tests for QPL retention (see paragraph 4.4.4).
2/ The number indicates the requirement or test specified in MIL-C-22442/1.

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TABLE I (Continued).

Inspection	Requirement Paragraph	Test Paragraph	Group I Qualification Tests	Group II Quality Conformance Test 100%	Group III Sample Testing Per 4.5.1
GROUP B					
Dielectric Withstanding Voltage	3.5.5	4.6.7	X		X
Gauging 1/	3.5.6	4.6.8	X		X
Pull Test 1/	3.5.7	4.6.9	X		X
Bend Relief Flexibility	3.5.10	4.6.12	X		X
GROUP C					
Low Temperature Flexibility	3.5.11	4.6.13	X		X
Durability 1/	3.5.13	4.6.15	X		X
Ozone Resistance 1/	3.5.14	4.6.16	X		X
Jacket Elongation	3.5.15	4.6.17	X		X
GROUP D 1/					
Temperature-Altitude 3/	3.6.1	4.7	X		
Salt Fog	3.6.2	4.7	X		
Vibration	3.6.5	4.7	X		
Shock	3.6.6	4.7	X		
Humidity	3.6.6	4.7	X		
Thermal Shock	3.6.6	4.7	X		

1/ Minimum tests for QPL retention (see paragraph 4.4.4).
3/ M22442/22-1 all altitude steps.
M22442/22-2 steps 3 and 12 only.

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